The Kidneys

The kidneys are two bean-shaped organs, each about the size of a fist, found on either side of the spine just below the rib cage. Each kidney has an outer renal cortex, an inner renal medulla made of renal pyramids, and a renal pelvis that collects urine and drains it into the ureter. Blood enters through the renal artery and leaves through the renal vein. The functional unit of the kidney is the nephron (about one million per kidney), made of a glomerulus that filters the blood and a tubule that adjusts the filtrate.

The functions of the kidneys:

- **Filtration and excretion:** removing urea, creatinine and excess salts from the blood.
- **Water and electrolyte balance:** controlling the levels of water, sodium and potassium.
- **Acid–base balance:** keeping the blood's pH at a normal level.
- **Blood pressure control:** releasing renin and regulating blood volume.
- **Hormone production:** erythropoietin (EPO) stimulates RBC production in the bone marrow, and the kidneys activate vitamin D.